

Encoder Capacity Is the Binding Constraint in Unified Multi-Task Ultrasound Landmark Detection

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Abstract. The FoundUS challenge requires a single model to localise anatomical landmarks across nine heterogeneous ultrasound tasks spanning fetal, obstetric and cardiac domains, with between two and twenty-two landmarks per image. Given a fixed multi-task heatmap architecture, which design axis actually governs accuracy? We report a controlled ablation over five alternative configurations of a smaller encoder, each of which leaves the hardest task (head circumference, HC) immovable at approximately 30 px mean radial error (MRE). A single change of encoder, from DINOv2-Small to DINOv3-Large adapted with LoRA (2.39 M of 305 M parameters trainable, 0.78%), reduces average MRE from 11.01 to 8.61 px and HC from 30.5 to 19.1 px. We additionally show that detecting and cropping split-screen acquisition artefacts in cardiac views yields order-of-magnitude gains (IVC 40.2 to 2.76 px; PSAX 27.4 to 1.70 px). Finally, we quantify a covariate shift between our held-out split and the challenge validation set, and show through a controlled distortion experiment that it is attributable to acquisition geometry unseen during training. We conclude that for unified multi-task ultrasound landmark detection, representational capacity – not loss or resolution engineering – is the binding constraint.

Keywords: Ultrasound · Landmark detection · Multi-task learning · Foundation models · Parameter-efficient fine-tuning

1 Introduction

Quantitative ultrasound biometry underpins routine clinical decision-making. Head circumference and biparietal diameter inform gestational age estimation; femur length supports growth assessment; angle of progression guides labour management; inferior vena cava calibre indicates volume status; and cardiac chamber dimensions are central to echocardiographic assessment. Each of these measurements is derived from a small number of anatomical landmarks, and each has historically been addressed by a dedicated, task-specific model.

The FoundUS challenge departs from this convention by requiring a *single unified model* to serve nine tasks at once. This is not merely an engineering

convenience. Unification changes the design problem: a loss weighting or output resolution that benefits one view may degrade another, and the practitioner is left without guidance on which axis of the design space deserves effort. Our aim in this work is to answer that question empirically.

We fix a standard multi-task heatmap architecture – a shared encoder feeding task-specific decoder heads – and vary one axis at a time. Our central observation is that five alternative configurations of the smaller encoder – spanning hyperparameter variation, foreground-weighted loss and increased heatmap resolution – leave the hardest task essentially unchanged, while a single substitution of the encoder resolves it. We report the negative results in full, because their consistency is what makes the positive result interpretable.

Our contributions are:

1. A controlled ablation over five configurations of a fixed architecture, all of which fail to move the hardest task (Section 4.1).
2. Evidence that encoder capacity is decisive: DINOv3-Large with LoRA breaks a plateau that five other configurations could not (Section 4.2).
3. A split-screen artefact detector and cropping procedure yielding order-of-magnitude improvements on cardiac views (Section 4.3).
4. A quantified analysis of the generalisation gap between our held-out split and the challenge validation set, with a controlled experiment isolating acquisition geometry as the cause (Section 4.4).

2 Related Work

Ultrasound landmark detection. Heatmap regression [5] remains the dominant paradigm for anatomical landmark localisation, in which each landmark is represented as a Gaussian-blurred target in a spatial output map and recovered by soft-argmax decoding. Prior work in ultrasound biometry has largely trained one model per view and measurement. Recent intrapartum biometry benchmarks [9] exemplify this task-specific convention.

Vision foundation models in medical imaging. Self-supervised vision transformers, notably the DINO family [3,1], produce dense features that transfer well to dense prediction tasks. Their utility in medical imaging is of particular interest given the persistent scarcity of expert annotation.

Parameter-efficient fine-tuning. Low-rank adaptation (LoRA) [2] injects trainable low-rank matrices into frozen attention projections. This matters here for a specific reason: full fine-tuning of a 300 M-parameter encoder on approximately 7K labelled images would be expected to overfit severely, and LoRA is what makes a large encoder viable in this regime. We did not ablate LoRA against full fine-tuning or alternative PEFT methods: at ~ 7 K labelled images full fine-tuning of a 305 M encoder overfits readily, and our claim concerns encoder *capacity* rather than the specific adaptation mechanism; disentangling the two is left to future work.

Concurrent evidence. A multi-task ultrasound challenge held at ISBI 2025, spanning classification, segmentation, regression and detection over a large-scale dataset, was won by an entry also employing DINOv3-Large with LoRA. We regard this convergence as independent corroboration of the claim advanced here.

3 Method

3.1 Problem setting

The challenge comprises nine tasks: A4C (16 landmarks), AOP (4), FA (4), FUGC (2), HC (4), IVC (2), PLAX (22), PSAX (4) and fetal femur (2). Training data comprises approximately 7,400 annotated images; the public validation set contains 619. Scoring weights mean radial error and clinical parameter error equally, aggregated in a rank-based, domain-balanced framework. The challenge and its baseline are described by Deng et al. [8].

3.2 Architecture

A shared encoder [6] produces a dense patch-token grid, which each task-specific head decodes to a 128×128 heatmap per landmark. Landmark coordinates are recovered by per-channel soft-argmax and normalised by the original image dimensions.

Input images are resized to 512×512 by letterboxing (longest side scaled to 512, remainder zero-padded symmetrically), preserving aspect ratio, followed by ImageNet normalisation. With a patch size of 16, this yields a 32×32 token grid.

Our final encoder is DINOv3-Large [1] (`vit_large_patch16_dinov3.lvd1689m`) adapted by LoRA of rank 16 applied to the attention query-key-value and projection matrices ($\alpha = 32$, dropout 0.05), giving 2.39 M trainable parameters out of 305 M (0.78%).

3.3 Split-screen artefact handling

A subset of cardiac acquisitions (A4C, PSAX, PLAX, IVC) are dual-panel screenshots in which the same anatomy appears twice, or in which a second modality panel (e.g. M-mode or Doppler) is displayed alongside the B-mode image. Annotations refer to only one panel. We detect these with a lightweight heuristic rather than a learned model: for each image we compute a column-wise mean-intensity profile, locate candidate vertical dividers as sharp low-intensity troughs, and score left-right symmetry of the two resulting panels by normalised cross-correlation. An image is classified as split-screen when a divider near the horizontal midline yields a symmetry score above a fixed threshold; the panel containing the annotated landmarks is then retained. On the labelled cardiac data the detector removed no valid single-panel image (zero false-crop rate on the training set), reflecting the conservative threshold; being training-time only, any residual error affects data curation rather than test-time inference.

Table 1. Ablation over six configurations of the DINOv2-Small architecture (rows 0–5), compared against two independent runs of the DINOv3-Large encoder (rows 6–7). Per-task mean radial error in pixels, lower is better. The small, high-contrast cardiac tasks (A4C, PLAX, PSAX, IVC) are already near their final values under the small encoder; only HC and fetal femur remain pinned high, and only the encoder substitution moves them.

#	Configuration	Avg	A4C	AOP	FA	FUGC	HC	IVC	PLAX	Femur
0	DINOv2-S baseline	11.01	3.74	6.24	10.25	8.77	30.48	4.93	3.26	29.60
1	hp variant	10.75	3.97	6.69	10.27	8.24	29.48	4.87	3.58	27.86
2	hp variant	11.17	4.19	6.22	10.44	9.04	31.34	4.12	3.30	30.09
3	hp variant	11.64	4.77	7.07	9.54	8.69	31.89	5.94	3.70	31.30
4	+ FG-weighted loss	11.43	4.46	6.24	10.77	9.19	30.11	7.01	3.43	29.89
5	+ heatmap res. 256	12.65	4.49	7.20	12.06	10.12	37.35	5.20	2.86	33.51
6	DINOv3-L + LoRA (A)	8.76	—	—	—	—	19.90	—	—	24.12
7	DINOv3-L + LoRA (B)	8.61	3.93	6.36	8.68	6.82	19.12	2.76	3.35	24.81

3.4 Training

Models are trained with Adam under a cosine schedule for 250 epochs at batch size 4 in bfloat16 precision, with photometric augmentation [7] (brightness, contrast, gamma, Gaussian noise). Model selection uses best validation average MRE. Training was performed on a single NVIDIA GH200 GPU (Isambard-AI), requiring approximately five minutes per epoch.

4 Experiments

4.1 Configurations that do not resolve the hardest task

Table 1 reports our ablation. The baseline uses a DINOv2-Small encoder with masked-autoencoder [4] pretraining on 164K unlabelled in-domain ultrasound images.

Three observations. First, HC remains between 29.5 and 37.4 px across every configuration of the smaller encoder; the best variant improves average MRE only marginally (11.01 to 10.75) while leaving HC essentially unchanged. Second, increasing heatmap resolution is actively harmful, degrading both average MRE and HC substantially. We attribute this to a mismatch between decoder resolution and encoder feature quality: a finer output grid demands finer spatial discrimination than the 32×32 token representation supports, yielding a harder optimisation target rather than a more precise one. Third, the two independent runs of the larger encoder (rows 6 and 7) agree closely on both average MRE (8.76, 8.61) and HC (19.90, 19.12), indicating the improvement is a property of the representation rather than of a favourable initialisation. The full per-task breakdown (Table 1) sharpens the picture: the small, high-contrast cardiac tasks are already solved by the small encoder and do not benefit materially from the

Table 2. Per-task mean radial error (pixels) on our held-out split. The encoder substitution reduces average MRE by 22% and HC by 37%.

Encoder	Avg	A4C	AOP	FA	FUGC	HC	IVC	PLAX	PSAX	Femur
DINOv2-S (baseline)	11.01	3.74	6.24	10.25	8.77	30.48	4.93	3.26	1.78	29.60
DINOv3-L + LoRA	8.61	3.93	6.36	8.68	6.82	19.12	2.76	3.35	1.70	24.81

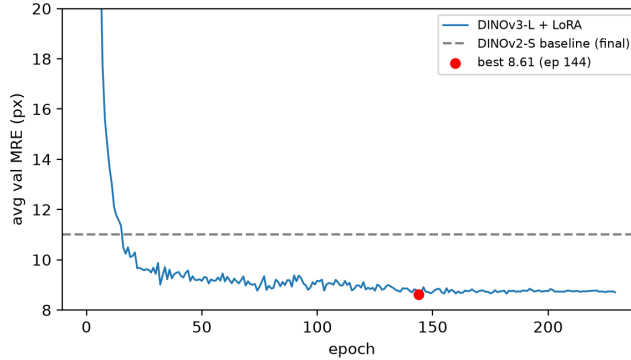


Fig. 1. Validation MRE per epoch for DINOv3-L + LoRA, against the final performance of the DINOv2-S baseline (dashed). The best checkpoint (8.61 px) is reached at epoch 144, after which performance plateaus.

substitution, whereas HC and fetal femur – the two tasks requiring integration of a large, low-contrast boundary – carry essentially the entire gain. The binding constraint is thus task-specific: it acts precisely where spatial-semantic capacity is required.

4.2 Encoder capacity resolves the plateau

Replacing the encoder with DINOv3-Large adapted by LoRA, holding all other components fixed, produces the results in Table 2.

The improvement is concentrated where it matters: HC, which resisted five prior configurations, falls by 37%. We emphasise that this is achieved with 0.78% of parameters trainable. The gain therefore cannot be attributed to additional fitting capacity in the task heads, which are unchanged; it follows from the richer frozen representation. Validation MRE converged at epoch 144 and remained flat between 8.61 and 8.81 through epoch 250 (Fig. 1).

4.3 Split-screen artefact removal

Table 3 reports the effect of the procedure in Section 3.3.

The magnitude of this effect – more than an order of magnitude on both tasks – deserves emphasis. In a multi-task setting where domains differ in acquisition

Table 3. Effect of split-screen detection and cropping on cardiac views.

Task	Without cropping	With cropping
IVC	40.2	2.76
PSAX	27.4	1.70

Table 4. Per-task performance on the challenge validation server for our submitted model (average MRE 25.60 px). MRE is mean radial error; MAE is the clinical-parameter error reported by the official scorer. The gap to our held-out split (Table 2) is concentrated in HC and PSAX, the tasks most affected by acquisition-geometry shift.

	A4C	AOP	FA	FUGC	HC	IVC	PLAX	PSAX	Femur
MRE	23.58	15.02	21.44	10.35	49.82	25.42	16.73	41.66	26.39
MAE	16.50	11.33	98.89	9.33	79.22	12.21	8.63	25.09	15.19

convention, per-domain data hygiene can dominate architectural tuning. We note that the organisers have confirmed the test set contains no split-screen images; the contribution is therefore to training-data quality rather than to test-time processing.

4.4 Generalisation gap and its cause

Our model attains 8.61 px on our held-out split but 25.6 px on the challenge validation server. We investigated this discrepancy systematically and report the outcome in full, as the diagnosis is informative.

We first verified that the coordinate mapping between the letterboxed input frame and original image pixels is correct, by evaluating predictions against ground-truth annotations in original pixel space (HC: 11.25 px), confirming no systematic transformation error. We then tested and rejected three candidate explanations: an inverse letterbox mapping at inference (which degraded the score to 48.5), aspect-ratio normalisation at inference (49.8 to 49.7, no material change), and the presence of uncropped split-screen images in the validation set (none are present).

The remaining explanation is covariate shift in acquisition geometry. Our HC training pool consists of a single image geometry (540×800 , aspect ratio 1.48). The challenge validation set spans five distinct geometries with aspect ratios from 1.45 to 1.90. To test whether this alone accounts for the degradation, we took training images with known ground truth and synthetically stretched them to validation-like aspect ratios (Table 5).

Error more than triples on identical anatomy under a geometry the model has not seen. This localises the failure precisely, and yields a caveat to our central claim: encoder capacity confers within-distribution accuracy, but it does not confer invariance to transformations the training data never exhibits.

Table 5. Controlled distortion experiment on HC training images with known ground truth. Error more than triples under an aspect ratio absent from training.

Condition	MRE (px)
Native aspect ratio (1.48, as in training)	10.5
Stretched to aspect ratio 1.90 (validation-like)	35.1

Mitigation attempts. Because Section 4.4 identifies acquisition geometry as the cause, we evaluated the two most direct remedies. First, geometric augmentation with keypoint tracking (independent-axis scaling 0.75–1.30, translation $\pm 5\%$, rotation $\pm 10^\circ$); second, stronger photometric augmentation (motion/Gaussian/median blur, multiplicative noise, and JPEG-compression artefacts) targeting appearance robustness. Both *improved* held-out MRE (to 7.69 and 7.52 px respectively, from 8.61) yet left challenge-server MRE unchanged or marginally worse (26.4 and 26.4 px, from 25.6). This dissociation is itself evidence for the covariate-shift diagnosis: augmentation that enriches the training distribution improves within-distribution accuracy but does not close a gap driven by geometries the training data cannot demonstrate. We therefore caution that augmentation alone is unlikely to resolve such gaps; multi-centre training data spanning the deployment geometries, or a validation split stratified by acquisition geometry so that local metrics track the target distribution, are the more promising directions.

5 Discussion

Why should capacity dominate? The tasks that resist intervention – HC and fetal femur – require integrating a large, low-contrast, frequently occluded boundary, in contrast to the small high-contrast structures of the cardiac views. Localising such boundaries demands both a wide effective receptive field and features that remain discriminative under speckle. Neither is obtainable by reweighting a loss or refining an output grid; both follow from a stronger representation.

This also explains the failure of increased heatmap resolution. Decoder resolution can only be exploited when encoder features support the corresponding spatial discrimination; beyond that point, additional output resolution adds optimisation difficulty without adding information.

Our results carry a practical implication. Given a fixed annotation budget and limited compute, effort is better spent on the strongest available pretrained encoder, adapted parameter-efficiently, than on loss or resolution engineering.

Scope of the claim. Our conclusion is bounded by the task regime. Capacity binds here because the decisive tasks require integrating large, low-contrast structures under speckle; where tasks are small and high-contrast (as with the cardiac views, already near-solved by the small encoder), capacity is slack and loss or resolution engineering may bind instead. We therefore state the claim conditionally: for unified multi-task suites containing at least one large-structure, low-contrast task, encoder capacity is the first axis to exhaust.

Limitations and future work. The analysis in Section 4.4 identifies geometric augmentation – aspect-ratio jitter, scale variation and random resized cropping with keypoint tracking – as the indicated remedy for the observed covariate shift; we report the outcome of evaluating this above (Section 4.4); it did not close the gap, redirecting future effort toward geometry-stratified splits and multi-centre data. A second direction follows from the central finding itself: if capacity is the binding constraint, larger backbones (e.g. ViT-H+/16 at 840 M parameters) merit evaluation. Finally, our parameter error analysis is incomplete for tasks whose clinical parameter is a composite measurement, and closing this gap requires the official per-task parameter definitions.

6 Conclusion

For unified multi-task ultrasound landmark detection, encoder capacity is the binding constraint. Five controlled configurations of a smaller encoder left the hardest task unchanged; a parameter-efficient encoder substitution resolved it. Combined with per-domain artefact handling, this reduced average MRE from 11.01 to 8.61 px and HC from 30.5 to 19.1 px. We further show that the residual gap to challenge validation performance is attributable to acquisition geometry unseen during training, delimiting what representational capacity alone can provide.

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